

Exp. No. 9	
Date:	

AIM:

ALGORITHM:

PROCEDURE:

Open Command Prompt as Administrator.

Navigate to the Snort directory:
cd C:\Snort

Verify Snort Installation:
Execute: bin\snort.exe -V

```
C:\Snort>bin\snort.exe -V

    ,,-
o"  )~
    '

-*> Snort! <*-
Version 2.9.20-WIN64 GRE (Build 82)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using PCRE version: 8.10 2010-06-25
Using ZLIB version: 1.2.11
```

Execute:bin\snort.exe -W

Index	Physical Address	IP Address	Device Name	Description
1	00:00:00:00:00:00	disabled	\Device\NPF_{11DB4899-83C2-4FE5-82EC-275F686931A}	WAN Miniport (Network Monitor)
2	disabled	disabled	\Device\NPF_{3B63C502-238A-45FA-A353-981DF66C46D}	WAN Miniport (IPv6)
3	00:00:00:00:00:00	disabled	\Device\NPF_{C8FC0DF66-700A-48CC-F06D-DC65F791A13A}	WAN Miniport (IP)
4	10:68:3E:7F:3F:45	169.254.16.222	\Device\NPF_{4980DC38-6F81-4108-8652-27F67C9392DB}	Bluetooth Device (Personal Area Network)
5	10:68:3E:7F:3F:45	169.254.16.156	\Device\NPF_{284CCE14-471E-4973-DC5E-3F2163572F82}	Realtek RTL8357BE WiFi 6 802.11ax PCIe Adapter
6	00:50:56:0C:00:08	192.168.32.1	\Device\NPF_{A41F7A70-3BBE-4D00-8CE3-B2B8E8F01457}	VMware Virtual Ethernet Adapter for VMnet8
7	00:50:56:0C:00:08	192.168.219.1	\Device\NPF_{1C872A44-DEBB-4D6C-A69E-1C98B330E1B1}	VMware Virtual Ethernet Adapter for VMnet1
8	16:68:3E:7F:3F:42	169.254.21.31	\Device\NPF_{794EBFFE-6036-4B8C-A3A4-39CAF8F729AF}	Microsoft Wi-Fi Direct Virtual Adapter #2
9	12:68:3E:7F:3F:42	169.254.115.114	\Device\NPF_{28AF3E9B-96F6-4FBB-A1E2-3BC9F8ACBB89}	Microsoft Wi-Fi Direct Virtual Adapter
10	00:00:00:00:00:00	0000:0000:0000:0000:0000:0000	\Device\NPF_{00000000-0000-0000-0000-00000000}	Loopback Pseudo-Interface for traffic capture
11	00:00:00:00:00:00	169.254.41.180	\Device\NPF_{11A679BF93-2490-49B8-81AB-EDD047590B9A}	ExpressVPN TUN Driver

[illegible]

Command: bin\snort.exe -vd -i 5

[illegible]

Command: bin\snort.exe -X -i 5

```
WARNING: No preprocessors configured for policy 0.  
09/05-13:47:09.571307 172.64.155.209:443 -> 192.168.0.156:63587  
TCP TTL:58 TOS:0x0 ID:9564 IpLen:20 DgmLen:40 DF  
****AP*** Seq:0x8584BAC Ack:0xA36565A8 Win:0x126 TcpLen: 20  
0x0000: 10 68 38 E7 3F 42 8C 86 DD B7 38 CC 08 00 45 00 .h8.?B....8...E..  
0x0010: 00 28 25 5C 40 00 3A 06 12 1E AC 40 9B D1 C0 A8 .(%)@.....@....  
0x0020: 00 9C 01 BB F8 63 08 58 4B AC A3 65 65 A8 50 10 .....c.XK..ee.P..  
0x0030: 01 26 4E 27 00 00 00 00 65 63 19 5B ..&N'....ec.[  
  
+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=  
  
WARNING: No preprocessors configured for policy 0.  
09/05-13:47:09.571308 172.64.155.209:443 -> 192.168.0.156:63587  
TCP TTL:58 TOS:0x0 ID:9565 IpLen:20 DgmLen:79 DF  
****AP*** Seq:0x8584BAC Ack:0xA36565A8 Win:0x126 TcpLen: 20  
0x0000: 10 68 38 E7 3F 42 8C 86 DD B7 38 CC 08 00 45 00 .h8.?B....8...E..  
0x0010: 00 04 25 5D 40 00 3A 06 11 F6 AC 40 9B D1 C0 A8 .0%[@.....@....  
0x0020: 00 9C 01 BB F8 63 08 58 4B AC A3 65 65 A8 50 18 .....c.XK..ee.P..  
0x0030: 01 26 51 34 00 00 17 03 03 00 22 D9 68 B8 56 A9 ..&Q4.....".h.v..  
0x0040: B4 82 50 A1 08 54 0D B8 B4 95 21 92 0E 7E EC 39 ...P.T.....l...r..  
0x0050: 75 88 DE F0 86 37 C7 CB 12 56 7F 72 E0 ....7...V.r..  
  
+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=+=+|=  
  
WARNING: No preprocessors configured for policy 0.  
09/05-13:47:09.614181 192.168.0.156:63587 -> 172.64.155.209:443  
TCP TTL:128 TOS:0x0 ID:57747 IpLen:20 DgmLen:40 DF  
****AP*** Seq:0xA36565BA Ack:0x8584BD3 Win:0x1FF TcpLen: 20  
0x0000: 8C 86 DD B7 38 CC 10 68 38 E7 3F 42 08 00 45 00 ....8..h8.?B..E..  
0x0010: 00 28 E1 93 40 00 80 06 0F E6 C0 40 08 9C AC 40 .(.@@.....@....  
0x0020: 9B D1 F8 63 01 BB A3 65 65 A8 08 58 4B D3 50 10 .....e.ee..XK.P..  
0x0030: 01 FF 4D 27 00 00 00 ..M'..
```

Packet Logging

First create a log directory if required:

```
mkdir log
```

Run: `bin\snort.exe -dev -l log -i 5`

Check the Log Folder

```
dir log
```

OUTPUT:

```
C:\Snort>dir log
Volume in drive C is Windows
Volume Serial Number is 749E-50E7

Directory of C:\Snort\log

05-09-2026  01:49 PM    <DIR>          .
05-09-2026  11:43 AM    <DIR>          ..
05-09-2026  01:49 PM                140 snort.log.1788596348
                1 File(s)                140 bytes
                2 Dir(s)  130,862,678,016 bytes free
```

Configure Snort for ICMP Detection

This is the main IDS demonstration.

Instead of depending on the full `snort.conf`, we can use a simple custom rule directly.

Navigate to the rules folder:

```
cd C:\Snort\rules
```

Create the rule file: `notepad local.rules`

When Notepad opens, enter:

```
alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
```

Save the file as:

```
local.rules
```

Understanding the IDS Rule

The rule is:

```
alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
```

Start Snort IDS

Return to the Snort directory:

```
cd C:\Snort
```

Execute: `bin\snort.exe -A console -q -i 5 -R rules\local.rules`

Keep this Command Prompt open.

Generate Test Traffic

Open a second Command Prompt.

Execute:

ping 8.8.8.8

IDS Alert Output

Return to the Command Prompt where Snort is running.

OUTPUT:

```
C:\Snort>bin\snort.exe -A console -q -i 4 -c rules\local.rules
09/03-18:41:00.537775  ** [1:1000001:1] ICMP Ping Detected ** [Priority: 0] {ICMP} 8.8.8.8 -> 192.168.1.2
09/03-18:41:01.553379  ** [1:1000001:1] ICMP Ping Detected ** [Priority: 0] {ICMP} 8.8.8.8 -> 192.168.1.2
09/03-18:41:02.577346  ** [1:1000001:1] ICMP Ping Detected ** [Priority: 0] {ICMP} 8.8.8.8 -> 192.168.1.2
09/03-18:41:03.601327  ** [1:1000001:1] ICMP Ping Detected ** [Priority: 0] {ICMP} 8.8.8.8 -> 192.168.1.2
```

RESULT: